

Anvi Agarkar

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anviagarkar.github.io · Dublin, Ireland · Stamp 1G Eligible – No Sponsorship Required

PROFILE

Final-year CS & Software Engineering student at Maynooth University, on track for first class honours by June 2026. Built a cardiac screening ML model, led frontend on a Microsoft-mentored passkey authentication system awarded 100%, and spent six months automating enterprise workflows at UNUM as a Pega CSA certified engineer. Stamp 1G eligible.

EDUCATION

BSc Computer Science & Software Engineering

Maynooth University, Ireland · Sept 2022 – Sept 2026

Year 3: 82% · Year 2: 75.3% · Year 1: 74%

Semester Abroad

Boston College, USA · Aug – Dec 2024

EXPERIENCE

Digital Process & Automation Intern · UNUM

Feb 2025 – Aug 2025

- Automated 10–12 business workflows using Pega and Constellation within a 10-person Agile team, streamlining internal processes and reducing manual effort; earned Pega CSA certification
- Delivered 10+ user stories end-to-end within Agile sprints including writing, unit testing, debugging and merging code on the Pega platform with senior developer review

Lab Demonstrator · Maynooth University

Oct 2025 – May 2026

- Support ~120 students across Theory of Computation, Databases, and Software Engineering & Processes by running lab sessions, marking work, and providing one-to-one guidance each term

PROJECTS

ECG Monitoring & SHD Detection — Final Year Project

Oct 2025 – May 2026

[Python](#) · [PyTorch](#) · [TensorFlow/Keras](#) · [Pandas](#) · [NumPy](#) · [ResNet-1D](#) · [PhysioNet](#)

- Adapted a pretrained multi-lead ResNet-1D for single-lead SHD detection using a custom lead-masking approach
- Achieved AUROC 0.74 / AUPRC 0.68 using Lead I vs. 12-lead baseline of 0.82 / 0.76, showing promising reduced-lead performance for low-resource and wearable cardiac screening

Password-less Biometrics Authentication — Microsoft (100%)

Feb – May 2024

[React](#) · [Node.js](#) · [Express](#) · [MongoDB](#) · [TypeScript](#) · [SimpleWebAuthn](#) · [Azure](#)

- Led frontend development in a 6-person team to deliver a full passkey authentication system supporting fingerprint and facial recognition across web and mobile;
- Mentored directly by Microsoft engineers through weekly design reviews; awarded 100%

SafarNama iOS App

Aug 2024 – Present

[Swift](#) · [SwiftUI](#) · [SwiftData](#) · [MapKit](#) · [Spring Boot](#) · [Java](#) · [PostgreSQL](#) · [Docker](#) · [JWT](#) · [Rest API](#)

- Built a full-stack travel diary iOS app with a custom Spring Boot backend - JWT authentication, bcrypt password hashing, and a REST API for cross-device sync
- Used SwiftUI and SwiftData for local persistence, MapKit for location search, and native camera integration; deployed the backend on Render with Docker and a managed PostgreSQL database

Intern Security Challenge — UNUM

May – Aug 2025

[Opengrep](#) · [Semgrep](#) · [Static Analysis](#) · [Application Security](#)

- Built custom open-source security tools using Semgrep and Opengrep to detect vulnerabilities in application code aiming to replace expensive third-party tools such as Veracode which cost \$100k+ annually in licensing fees
- Presented the final solution to 300+ UNUM employees

SKILLS

Languages: Python, Java, JavaScript, TypeScript, HTML/CSS, SQL, Swift

ML / Data: PyTorch, scikit-learn, Pandas, NumPy, Time-Series Analysis, TensorFlow/Keras

Technologies: React, Node.js, Express, Spring Boot, MongoDB, Docker, Azure, Firebase, GitHub, REST APIs

Tools & Practices: Pega, Semgrep, OpenGrep, Git, CI/CD, Agile/Scrum, Unit Testing, Static Analysis

LEADERSHIP & ACHIEVEMENTS

- Academic Representative · Treasurer, Women in STEM · CS Centre Tutor